

A Reproducible Pipeline for JunoCam Limb Analysis and the Detection of Detached Jovian Hazes as a Function of Planetocentric Latitude

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Date: June 16, 2026

Abstract

We present a reproducible, directory-agnostic, stage-aligned pipeline for detecting detached-haze candidates in Jupiter's atmosphere from JunoCam limb observations. The workflow processes each image programmatically from IMG ingestion and SPICE attachment through limb tracing, geometric rectification, perpendicular brightness sampling, peak classification, and fragment-level aggregation. A geometry-conditioned detectability criterion distinguishes interpretable non-detections from resolution-limited fragments by relating slant distance to an empirical limb-resolution threshold. The revised pipeline further distinguishes geometry-valid non-detections from fragments lacking physically interpretable limb geometry, preventing sparse or geometry-limited regions from being misinterpreted as physical absence of detached haze. Applied across Perijove 14, the detectability-screened aggregate reveals a pronounced mid-latitude maximum in detached-haze candidate occurrence centered near $\sim 33^\circ$ planetocentric latitude, with substantially lower occurrence at equatorial and higher latitudes. Longitude–latitude aggregation shows structured clustering while preserving explicit coverage limitations. Validation analyses indicate that the observed enhancement is unlikely to be explained solely by observational geometry and is consistent with a real atmospheric distribution.

Introduction

Jupiter's upper-atmospheric haze plays an important role in the planet's radiative balance, photochemistry, and large-scale circulation. Detached haze layers—optically thin scattering features residing above the main cloud deck—have been observed intermittently in limb-viewing geometry and are thought to reflect latitude-dependent atmospheric processes. However, their occurrence and spatial distribution remain difficult to quantify due to the geometric constraints of limb observations and the complexity of spacecraft imaging data.

NASA's Juno spacecraft provides repeated high-resolution limb views of Jupiter through its visible-light camera, JunoCam, across many perijove passes. These observations contain information about vertical atmospheric structure, but extracting that information reliably requires accurate geometric context, consistent sampling geometry, and careful handling of cases where limb crossings are ambiguous or poorly constrained.

Previous work (Kelly 2022) demonstrated that detached haze layers could be identified in JunoCam data by combining limb tracing with SPICE-derived geometry and brightness-profile inspection. While effective for exploratory analysis, such approaches are difficult to

reproduce or scale and rely heavily on manual intervention. The present work replaces that workflow with a fully programmatic, stage-aligned pipeline that makes each analytical decision explicit, reproducible, and independently auditable.

This study introduces a modular, directory-agnostic JunoCam limb-analysis pipeline designed to produce conservative measurements of detached haze detectability as a function of planetocentric latitude. The pipeline prioritizes geometric integrity and statistical robustness over aggressive feature localization, preserving atmospheric structure while avoiding over-interpretation in regions where SPICE geometry is undefined. By aggregating fragment-level detections statistically across images rather than over-localizing individual features, the method enables robust comparison across framelets and imaging geometries while remaining explicit about the limitations imposed by limb viewing.

Methods: Pipeline Overview

All processing and analysis stages are executed on a per-IMG basis. For a given JunoCam image, all framelets, limb fragments, diagnostic plots, and statistical summaries are generated within IMG-specific directories. This design ensures full reproducibility, prevents output collisions when processing multiple images, and allows cross-image aggregation to be performed as a distinct analytical step outside the scope of this study. Stages 1–7 operate on a single JunoCam image and define a complete, reproducible analysis pipeline; in Stage 8, this identical pipeline is executed independently on all images from a single perijove, and the resulting fragment-level outputs are aggregated in a distinct Perijove-scale analysis. Processing is performed via two resumable shell-based batch runners: one orchestrating Stages 01–02 and a second orchestrating Stages 03–07. These runners iterate deterministically across IMG directories while preserving IMG-scoped outputs, logs, and stage isolation. Existing outputs are detected and skipped automatically, enabling interruption-safe reruns and selective recovery of failed IMG products without recomputing the full Perijove.

The haze classification pipeline consists of a sequence of modular Python stages, each of which consumes the outputs of the preceding stage while remaining independently executable for diagnostic or exploratory analysis.

Pipeline Structure and Stage Alignment

The pipeline is explicitly organized into numbered stages (Stage 01–Stage 08), each corresponding to exactly one Python module. No stage relies on logic described in later stages, and no section contains forward references. This structure is intentional and enforces reproducibility, traceability, and modular reuse.

Stage	Module	Description	Primary Output
01	framelets_and_spice.py	IMG ingestion, framelet generation, SPICE attachment	SPICE-initialized .CUB framelets
02	trace_limb_polyline.py	Limb detection and tracing	Limb polylines, overlays
03	rectify_limb.py	Limb rectification and coordinate mapping	Rectified strips, mapping tables
04	plot_rectified_perpendiculars.py	Perpendicular sample extraction and SPICE annotation	Per-sample profile tables, overlays
04b	slant_distance_detectability.py	Fragment-level detectability screening	Detectability tables, diagnostics

05	graph_profiles.py	Diagnostic profile visualization	QC plots
06	peakfinder.py	Peak finding and fragment classification	Classified fragment tables, diagnostics
07	haze_analysis.py	Single-image latitude-binned occurrence analysis	Fragment table, binned occurrence outputs
08	perijove_analysis.py	Perijove-scale aggregation	Aggregate latitude/longitude occurrence outputs
08b	peak_to_peak_distance.py	Mean distance between peaks	Summary diagram

Note well: All terminal commands are executed from the project's root directory.

Stage 0 — Computational Environment (Assumed)

This pipeline assumes a working installation of ISIS3 and locally available SPICE kernels. Installation, operating-system configuration, and kernel acquisition are treated as prerequisites and are not addressed here.

To ensure numerical and geometric reproducibility, we provide a fully pinned conda environment named **JHC (Juno Haze Classifier)** that reproduces the exact computational context used in this study, including Python version, third-party libraries, and the ISIS runtime bindings.

All SPICE geometry is resolved using locally synchronized kernels rather than remote services, ensuring deterministic limb geometry and reproducibility independent of external servers. The specific kernels attached to each image cube are recorded verbatim in per-cube manifest files, providing a complete provenance record for every geometric solution used in downstream analysis. From this point forward, all pipeline commands are executed within the activated JHC environment in the project root directory. No further environment configuration occurs within the pipeline itself. All scientific decisions occur downstream of this assumption.

An additional Jupiter body-code kernel was supplied via the SPICE extra option to ensure consistent body-ID resolution during offline processing.

This pipeline is directory-agnostic; however, once an IMG root directory is defined, this internal structure is assumed. All stages read only from their preceding stage and write exclusively to IMG-scoped output directories.

Stage 1 — Data Ingestion and SPICE Initialization

1.1 Data Acquisition from the Planetary Data System

Raw JunoCam images are publicly available through NASA's Planetary Data System (PDS). Each image is distributed as an .IMG file accompanied by a corresponding .LBL 3metadata label, which contains spacecraft position, timing, and geometric information required for ISIS ingestion.

In this study, the pipeline follows Brueshaber (2021) and is demonstrated using JunoCam image JNCR_2018197_14C00024_V02.IMG, acquired during Perijove 14 on 16 July 2018. This image captures Jupiter's northern limb and serves as a concrete, reproducible example throughout all subsequent stages.

During Perijove-scale processing, IMG/LBL products were selected from multiple PDS archive volumes using the highest available product version for each observation. Because JunoCam archive products are distributed across mixed archive generations and non-contiguous numbering sequences, the pipeline resolves the most recent available IMG/LBL pair deterministically prior to Stage 01 ingestion. This prevents incomplete Perijove coverage arising from archive-version inconsistencies and ensures that downstream aggregation operates on the most recent publicly available calibrated products.

1.2 Ingestion into ISIS

Once the raw data are available locally, each .IMG file is ingested into ISIS as a .CUB, and full SPICE geometry is attached to the resulting image products. In this pipeline, ingestion is performed by executing `01.framelets_and_spice.py`, which processes each JunoCam .IMG/.LBL pair, breaks the raw image into its constituent framelets, and attaches locally cached SPICE kernels to each framelet cube. For reproducibility, the ingestion step records a per-framelet manifest that enumerates every SPICE kernel attached to that cube, ensuring that the complete geometric context of each product is preserved and auditable.

1.3 Execution

To execute Stage 1, run the following command from the environment's root directory:

```
python src/01.framelets_and_spice.py \
  --img data/JNCR_2018197_14C00024_V02/raw/JNCR_2018197_14C00024_V02.IMG \
  --outdir data/JNCR_2018197_14C00024_V02/cub/stage_01_framelets
```

Successful execution produces a set of SPICE-initialized .cub framelets and accompanying manifest files under `stage_01_framelets/`. Once these outputs are present, the pipeline proceeds to Stage 2.

Stage 2 — Limb Detection and Tracing

With SPICE geometry successfully attached, the next stage identifies Jupiter's planetary limb in each JunoCam framelet. The limb is defined as the boundary separating Jupiter's illuminated atmosphere from surrounding space and represents the geometric foundation for all subsequent limb-normal sampling and brightness-gradient analysis.

Limb detection is performed by the module `02.trace_limb_polyline.py`, which combines SPICE-derived camera geometry with image-based edge detection to produce a continuous, subpixel-resolution polyline describing the curvature of the visible atmospheric boundary in each framelet. Each SPICE-attached framelet (.cub) is exported as a 16-bit TIFF (science fidelity) and an 8-bit, linearly stretched PNG (for quick-look reference). The gradient-based search seeds a starting limb-point and iteratively proposes subsequent points along the edge by maximizing brightness differential across a small angular neighborhood. The resulting (x,y) pixel coordinates of the polyline are saved as `<cube> _LIMBENDPOINTS.csv`, while an overlay with cyan limb points is saved as `<cube> _OVERLAY.png` in the same directory.

The seeding step no longer assumes a fixed limb location. Instead, the algorithm identifies the global maximum of the Sobel gradient magnitude within each framelet, providing orbit-agnostic initialization and ensuring compatibility with any JunoCam observation geometry. This update replaces the earlier orientation-specific heuristic (cf. Wen, 2021) and improves cross-orbit generalization, detecting both upper and lower limb fragments even in oblique lighting geometries.

2.1 Processing Scope and Assumptions

All limb tracing and rectification steps are performed on a per-framelet basis under fixed geometric assumptions. Sampling density, smoothing parameters, and perpendicular construction are held constant across all framelets within a given IMG to ensure comparability. No framelet-level tuning is performed, and all parameters are documented explicitly to support reproducibility.

2.2 Execution

```
python src/02.trace_limb_polyline.py \  
  --cubdir data/JNCR_2018197_14C00024_V02/cub/stage_01_framelets \  
  --outdir data/JNCR_2018197_14C00024_V02/cub/stage_02_trace_polyline \  
  --noshow
```

Executing Stage 2 automates limb detection across SPICE-initialized framelets and writes the resulting limb polylines and verification overlays to IMG-scoped output directories. These artifacts constitute the sole geometric inputs to the limb-rectification process performed in Stage 3 and define the valid sampling boundary for all subsequent analysis.



Figure 2.2: Limb Detection Across Successive Framelets

2.3 Mandatory Manual Cleaning (Non-Optional)

Certain JunoCam framelets do not contain usable limb–space transitions, either because Jupiter fully occupies the frame or because image boundaries are dominated by blur, compression artifacts, noise, or ambiguous edge structure. These framelets must be excluded prior to downstream rectification and brightness-profile analysis.

Importantly, this manual intervention is not treated as a pipeline failure, but rather as a scientifically necessary quality-control stage within a broader semi-automated workflow. The revised pipeline distinguishes between (1) geometry-valid non-detections, which remain scientifically meaningful and are retained for downstream occurrence statistics, and (2) framelets lacking a physically interpretable limb–space transition, which must be removed to prevent artificial inflation of null detections and suppression of true atmospheric structure.

Figure 2.3 illustrates a representative false-positive case in which automated edge tracing identifies non-physical structure away from the limb. These exclusions are deliberate scientific curation decisions rather than algorithmic failures. Following manual QC, the retained Stage 02 limb traces are frozen into explicit manifest tables defining the accepted fragment population for downstream processing. These manifests distinguish intentionally excluded fragments from incomplete processing failures and ensure deterministic reproducibility during resumable reruns and interruption recovery.

Additional validation scripts are applied after QC to remove orphaned Stage 02 products (for example, limb-trace CSV files lacking corresponding overlay images), preventing partial reruns or interrupted processing from silently altering the retained fragment population. Portions of this user intervention are expected to be supported by future software automation, but manual QC is mandatory for the current implementation.

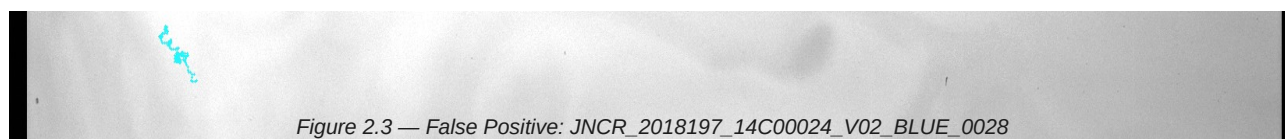


Figure 2.3 — False Positive: JNCR_2018197_14C00024_V02_BLUE_0028

Stage 3 — Limb Rectification and Perpendicular Sampling Geometry

The limb traces produced in Stage 2 define the precise two-dimensional boundary between Jupiter’s illuminated atmosphere and surrounding space. However, the curvature of this boundary varies across each framelet, making direct comparison of brightness profiles difficult in raw image coordinates.

Stage 3 transforms each detected limb segment into a limb-aligned coordinate system in which the curved atmospheric boundary is straightened and placed into a common reference frame. This rectification enables consistent, repeatable sampling of brightness variations perpendicular to the limb, corresponding to changes in scattering with altitude.

3.1 Limb-Aligned Rectification

Each framelet's limb polyline is resampled at regular intervals and locally approximated as a smooth curve. Using this curve, the original image is re-projected into a rectified coordinate system in which the limb appears as a straight horizontal feature (Figure 3.1). In this rectified image, horizontal position corresponds to distance along the limb, while vertical position corresponds to distance perpendicular to the atmospheric boundary. Rectification is performed independently for each framelet and restricted to the stable limb region, excluding end segments where the tangent orientation becomes ill-defined near image boundaries. This resulting rectified product is a narrow strip centered on the limb that preserves the local photometric structure across the atmosphere-space interface. The limb-aligned rectification strategy employed here follows the general methodology described by Eichstädt (2020), who introduced image-space rectification techniques for analyzing Jupiter's limb structure in JunoCam observations.

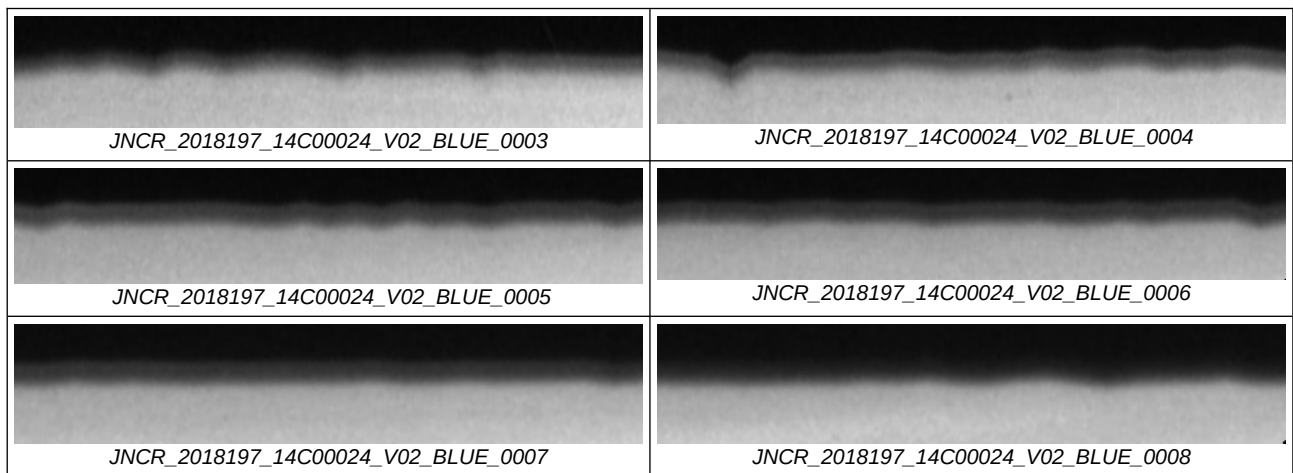


Figure 3.1 illustrates the geometric rectification process that transforms the curved limb into a flat, horizontally aligned reference frame.

3.2 Reversible Mapping Between Rectified & Original Coordinates

The rectification process is implemented as a fully reversible geometric transformation. For every pixel in the rectified image, the corresponding pixel coordinates in the original ISIS-calibrated framelet are recorded and stored in an explicit mapping table.

This rectified-to-original coordinate map ensures that any measurement performed in rectified space can be traced directly back to the original image pixels and associated SPICE geometry. As a result, no scientific interpretation relies solely on transformed imagery; all derived quantities remain physically anchored to the source observation.

For each framelet, Stage 3 produces:

- a 16-bit rectified limb image (*_RECTIFIED.tif) used for quantitative sampling,
- an 8-bit visualization image (*_RECTIFIED.png) used for diagnostics and figures
- a dense rectified-to-original coordinate mapping table (*_RECTIFIED_MAPPING.csv).

These products form the terminal outputs of Stage 3 and serve as requisite inputs for brightness-profile analysis downstream.

3.3 Execution

Stage 3 performs limb-aligned rectification for all validated limb traces produced in Stage 2, transforming each curved limb segment into a straightened reference frame centered on the atmospheric boundary (Figure 3.1). This step establishes the geometric basis required for all downstream brightness sampling and analysis.

To execute, run the following command from the project root directory:

```
python src/03.rectify_limb.py \  
  --indir data/JNCR_2018197_14C00024_V02/cub/stage_02_trace_polyline \  
  --imagedir data/JNCR_2018197_14C00024_V02/cub/stage_01_framelets \  
  --outdir data/JNCR_2018197_14C00024_V02/cub/stage_03_rectified
```

Users should verify that the rectified limb appears horizontally aligned and centered within each output image before proceeding. If rectified outputs are absent or malformed, downstream analysis must not be attempted. Formal validity criteria and automated quality-control procedures for intermediate rectified products are subjects of future work.

Stage 4 — Perpendicular Sampling, Brightness Profiles

4.1 Perpendicular Brightness Sampling Geometry

With the limb geometry straightened, sampling paths perpendicular to the atmospheric boundary are defined directly in rectified space. Sampling lines are placed at uniform horizontal intervals along the rectified limb and extend vertically across the limb transition, probing from the dark background of space into Jupiter's upper atmosphere.

For each sampling position, two limb edges are identified in rectified coordinates, and their midpoint defines the local limb's centerline. From this midpoint, a vertical sampling path is constructed that spans a fixed number of pixels both outward into space and inward toward the planet. These rectified limb fragments are parallel, evenly spaced, and orthogonal to the limb by construction, independent of the original viewing geometry.

Each sampled rectified pixel is mapped back to its original framelet coordinates using the rectified-to-original mapping table, and its corresponding 16-bit pixel value is retrieved from the ISIS-calibrated TIFF image. For each framelet, a single consolidated table is produced containing rectified coordinates, original image coordinates, and pixel intensity values for every sampled location. In addition to these numerical outputs, a diagnostic overlay image is generated for each rectified framelet (Figure 4.1), displaying the rectified limb strip with detected limb edges and perpendicular sampling paths superimposed.

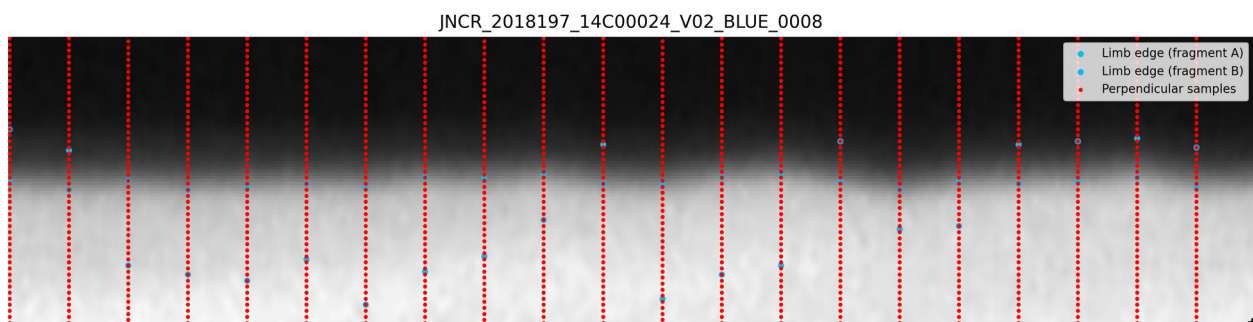


Figure 4.1. Detected limb edges (blue) along brightness sampling paths (red).

4.2 Association of Perpendicular Samples with SPICE Geometry

To associate each sampled pixel with physical planetary coordinates, spacecraft geometry is retrieved using the SPICE kernel system through the ISIS campt tool. For each sampled pixel location, SPICE is queried using the original unrectified image geometry, ensuring that physical coordinates are derived directly from calibrated spacecraft pointing and timing information.

For every sampling point along each perpendicular brightness profile, ancillary metadata—including planetocentric latitude, System III longitude, spacecraft altitude, emission angle, solar incidence angle, phase angle, and slant distance—are retrieved and appended to the sampling records. These parameters are stored alongside the pixel brightness values without altering the sampling order established in rectified space.

This approach decouples geometric sampling from physical interpretation: brightness profiles are constructed and ordered according to their limb-normal distance in rectified image coordinates, while each sample is independently associated with its original corresponding physical location on Jupiter. As a result, brightness-gradient analysis and peak detection are performed in a stable geometric coordinate system, while all reported results can be expressed in physically meaningful planetary coordinates.

4.3 Execution

To execute Stage 4, run the following command from the project's root directory:

```
python src/04.plot_rectified_perpendiculars.py \  
  --indir data/JNCR_2018197_14C00024_V02/cub/stage_03_rectified \  
  --framelet-dir data/JNCR_2018197_14C00024_V02/cub/stage_01_framelets \  
  --outdir data/JNCR_2018197_14C00024_V02/cub/stage_04_perp_samples \  
  --batch
```

Stage 4b — Slant-Distance Detectability Screening

Stage 04b converts the sample-level SPICE geometry produced in Stage 4 into fragment-level detectability metrics. Using the slant distance and spacecraft geometry associated with each limb fragment, the stage computes the effective limb resolution (km per pixel), the number of pixels subtended by a nominal detached-haze separation, and a corresponding detectability classification indicating whether detached haze would have been geometrically resolvable under the observed viewing conditions.

These detectability metrics are propagated into subsequent stages and enforced during occurrence aggregation so that detached-haze occurrence is evaluated only within the retained detectability-valid fragment population.

Stage 04b produces:

- a fragment-level detectability table,
- an image-level summary table,
- and two diagnostic figures illustrating detectability behavior as a function of latitude and slant distance.

4b.1 Geometry-Conditioned Detectability and the Eichstädt Criterion

Detached haze signatures are identified through vertical structure in limb-normal brightness-gradient profiles. Consequently, JunoCam's ability to resolve detached haze depends not only on atmospheric structure, but also on observing geometry. In particular, the effective vertical resolution at the limb degrades systematically with increasing slant distance between the spacecraft and the tangent limb.

Stage 04b implements a geometry-conditioned interpretability screen that distinguishes:

- **detectability-valid fragments**, in which detached haze would have been geometrically resolvable under the adopted criterion, from
- **geometry-limited fragments**, in which detached haze could not be reliably interpreted due to insufficient spatial resolution.

Importantly, this stage does not alter peak-finding thresholds or classification logic. Instead, it converts SPICE-derived geometry into fragment-level detectability metadata that are carried forward into downstream classification and aggregation stages.

Following Eichstädt's geometric argument, the slant distance to the tangent limb may be approximated by:

$$d^2 = 2ra + a^2$$

which provides a simple geometric relation between spacecraft altitude and line-of-sight distance to the tangent limb. In practice, the Stage 4 SPICE-derived slant distance values are treated as the primary source of truth, while the spherical expression above is retained as a consistency check and explanatory relation. JunoCam's horizontal field of view is 58° across 1600 detector pixels, corresponding to an angular scale of approximately 0.000633 radians per pixel. For small angles, the vertical spatial scale at the limb is therefore approximated by

$$km / pixel \approx d \theta_{px}$$

where d is slant distance in kilometers and θ_{px} is the JunoCam angular pixel scale in radians per pixel. This converts slant distance directly into an approximate limb resolution in km per pixel.

Using well-resolved detached-haze profiles at low slant distance as an empirical reference case, a nominal detached-haze separation of 76 km was adopted. Under favorable geometry, this separation corresponds to approximately 5 pixels at a slant distance near 24,000 km. This value is treated as an empirical interpretability benchmark rather than a fixed atmospheric constant.

For each fragment, the nominal detached-haze separation is converted into pixel units:

$$\text{pixels per detachment} = 76 \text{ km} / (\text{km} / \text{pixel})$$

Fragments are classified as detectability-valid only if: $N_{px} \geq 2.5$.

This defines the adopted Eichstädt-style criterion used throughout the revised pipeline:
geometry → pixel scale → resolvability → inclusion for interpretation.

Because Stage 4 sample rows may contain leading positions lacking finite SPICE geometry, fragment-level slant distance, latitude, and altitude are represented using the first finite SPICE-resolved value within each fragment. Median finite values are retained separately for auditability and cross-checking. No interpolation is introduced across SPICE-undefined regions.

The resulting fragment-level detectability structure is summarized in Figure 4b. Panel A shows pixels per nominal detached-haze separation as a function of latitude, while Panel B illustrates the dependence of limb resolution on slant distance.

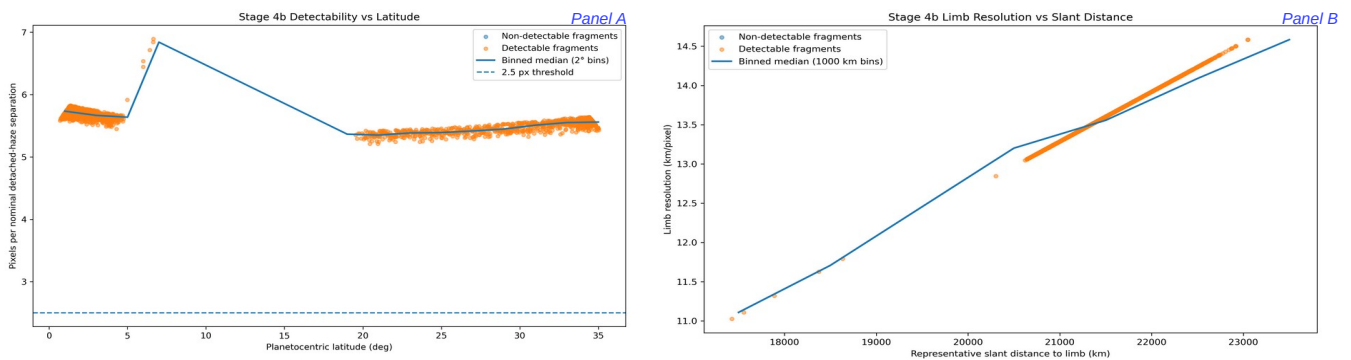


Figure 4b. Detectability diagnostics. Panel A shows fragment-level pixels per nominal detached-haze separation (76 km) as a function of planetocentric latitude for the processed .cub, with fragments classified as detectability-valid or geometry-limited relative to the adopted 2.5-pixel interpretability threshold. Panel B shows limb resolution (km per pixel) as a function of representative slant distance to the limb. These diagnostics separate geometry-limited fragments from geometry-valid non-detections prior to detached-haze occurrence aggregation and establish the retained interpretability structure inherited by downstream stages.

4b.2 – Execution

Stage 04b computes fragment-level detectability metrics from the SPICE-annotated perpendicular-sample tables generated in Stage 4 and writes both tabular and diagnostic outputs for downstream analysis.

```
python src/04b.slant_distance_detectability.py \
  --indir data/JNCR_2018197_14C00024_V02/cub/stage_04_perp_samples \
  --outdir data/JNCR_2018197_14C00024_V02/analysis/stage_04b_detectability
```

Successful execution produces:

- (1) a fragment-level detectability table (`stage_04b_fragment_detectability.csv`),

- (2) an image-level summary table (`stage_04b_img_summary.csv`), and
- (3) two diagnostic figures illustrating resolution and detectability behavior.

Stage 5 — Brightness-Gradient Visualization

Stage 5 is a diagnostic visualization stage. It does not create a required dependency for downstream detection. Profiles are visualized for validation only; all classification and detection logic is performed in Stage 6.

5.1 Construction of Limb-Normal Brightness-Gradient Profiles

For each rectified limb fragment produced in Stage 4, brightness is sampled along paths oriented perpendicular to the limb, yielding one-dimensional brightness profiles ordered by limb-normal distance (dy). These profiles probe the vertical structure of Jupiter's upper atmosphere, extending from optically thin space, across the tangent limb, and into the upper cloud deck. Where the PRIMARY peak itself lacks finite SPICE latitude, dy -space ordering is preserved and no interpolation is introduced at the detected haze peak.

Raw pixel intensities are extracted from calibrated ISIS products and assembled into ordered brightness sequences. To suppress slowly varying background illumination and emphasize vertical structure, first-order brightness gradients ($\Delta\text{Brightness}$) are computed along each profile as a diagnostic representation of vertical scattering structure. This transformation sharpens transitions associated with atmospheric boundaries and highlights secondary scattering layers above the cloud deck.

Where available, SPICE-derived physical coordinates (planetocentric latitude, longitude, slant distance, emission angle) are attached to each sampling point without altering the dy -based ordering.

For visualization purposes only, profiles may be plotted with the horizontal axis labeled in planetocentric latitude where SPICE geometry is available. Importantly, this relabeling does not alter the underlying index-space ordering used for peak detection or classification. The plotted latitude axis therefore serves as a physical annotation rather than a computational coordinate.

5.2 Smoothing and Noise Suppression

Brightness-gradient profiles often exhibit pixel-scale noise arising from photon statistics, local albedo variations, and residual detector artifacts. To suppress this noise while preserving physically meaningful extrema, all $\Delta\text{Brightness}$ profiles are smoothed using a Savitzky–Golay filter (quadratic polynomial, window size of 5 samples). The smoothing parameters are fixed and applied uniformly across all profiles. No adaptive or profile-specific tuning is performed.

Both raw and smoothed profiles are retained. Raw profiles are preserved for archival integrity, sanity checks, and archival transparency, while smoothed profiles are used exclusively for peak detection and classification. Because detached haze signatures manifest as coherent extrema across multiple samples, the applied smoothing suppresses

pixel-scale noise without erasing physically meaningful secondary structure.

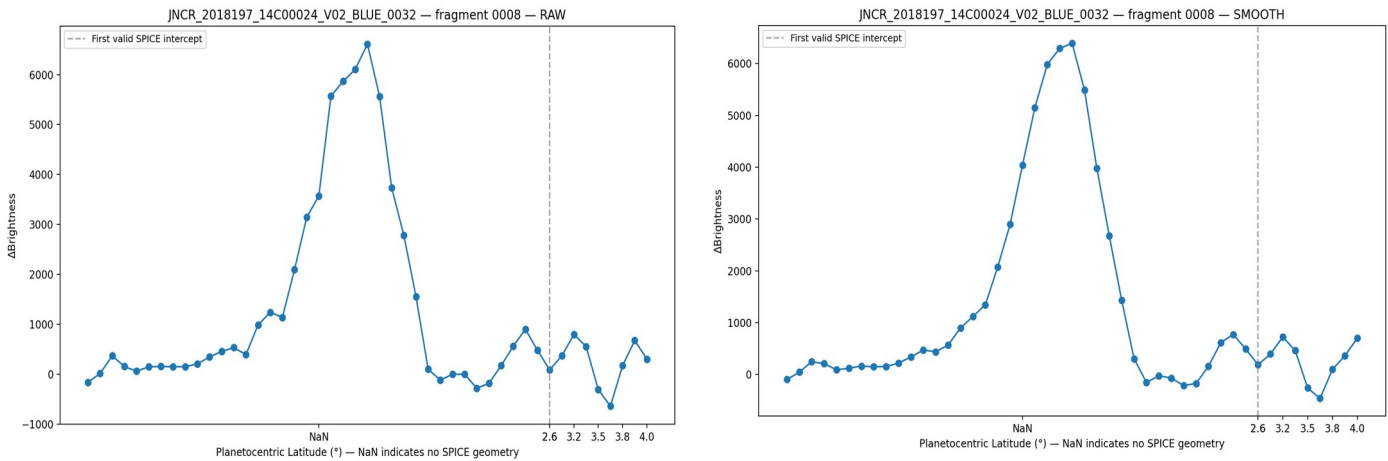


Figure 5.2 — Raw & Smoothed Δ Brightness Profiles illustrating noise suppression along the Jovian limb

5.3 Execution

To execute Stage 5, run the following command from the project root directory:

```
python src/05.graph_profiles.py \
  --indir data/JNCR_2018197_14C00024_V02/cub/stage_04_perp_samples \
  --outdir data/JNCR_2018197_14C00024_V02/cub/stage_05_graphs \
  --batch \
  --smooth 5
```

Stage 6 — Gradient Analysis and Peak-Finding

Stage 6 replaces earlier visualization-first workflows with a unified, physics-aware peak-detection system implemented in a single module (`06.peakfinder.py`) applied uniformly across all limb fragments. Diagnostic plots are generated directly from the same logic that performs classification, ensuring that all visualizations reflect the true decision path of the algorithm.

6.1 Peak Detection Logic and Physical Constraints

All profiles are analyzed strictly in dy-ordered space, preserving uniform sampling independent of SPICE coordinate availability. Stage 6 operates on smoothed Δ Brightness (dy) profiles computed directly from the Stage 4 perpendicular-sample tables. Smoothing is applied internally using the same fixed parameters described in Stage 5. Automated peak detection is performed on the smoothed Δ Brightness (dy) profiles using a constrained local-extrema search. The algorithm identifies candidate local maxima and classifies them according to physically motivated rules rather than purely mathematical prominence.

The dominant limb peak—corresponding to the sharp brightness transition at the

atmospheric boundary—is identified as the *primary* peak. Candidate secondary peaks are only evaluated as potential detached haze signatures if they satisfy the following constraints:

1. **Outboard Requirement:** Detached haze features must occur on the spaceward (outboard) side of the primary limb peak in limb-normal (dy) sampling. Any extrema occurring at or interior to the primary limb peak are excluded. This is a known and defensible geometric limitation inherent to limb-normal sampling and is consistent with the physical expectation that detached hazes reside above the cloud deck, between the cloud tops and the tangent limb.
2. **Separation Constraints:** Secondary peaks must be separated from the primary peak by a minimum and maximum distance in dy index space, eliminating spurious noise near the limb transition and irrelevant interior structure.
3. **Near-Limb Atmospheric Constraint:** Secondary candidates must lie within the outer portion of the atmospheric column sampled. Features occurring too deep in the interior are rejected as unlikely to represent detached haze.
4. **Absolute and Relative Brightness Thresholds:** Candidate secondary peaks must exceed both an absolute Δ Brightness floor and a fractional threshold relative to the primary peak. This combination suppresses low-amplitude noise while retaining physically plausible weak haze layers.
5. **Valley (Shoulder) Test:** To prevent misclassification of post-limb decay shoulders or monotonic slopes, each secondary candidate must exhibit a sufficient local valley between itself and the primary peak. This allows the algorithm to accept both resolved secondary maxima and physically meaningful shoulder-like features while rejecting simple brightness decay.

Among candidates that pass all constraints, the strongest secondary feature is selected as the detached haze indicator for that limb fragment.

6.2 Geometric Limitations, SPICE Association, and Detectability Provenance

A critical geometric limitation arises near the tangent limb: not all detected atmospheric structure corresponds to a unique or stable intercept with Jupiter's reference shape model. In these regions, SPICE geometry cannot assign a reliable planetocentric latitude, even though optically detectable haze structure is present.

As a result, neither the primary nor any secondary peak is guaranteed to coincide with a sample row possessing finite SPICE latitude. This is an inherent consequence of limb-viewing geometry, in which the line of sight intersects extended atmospheric structure without intersecting the modeled planetary surface. Accordingly, when finite SPICE geometry is unavailable exactly at the detected PRIMARY peak, the pipeline preserves the dy-space optical detection and, in later aggregation stages, uses the nearest finite SPICE-resolved row only for limb-crossing geographic assignment rather than altering the detection itself.

To preserve scientific integrity, the pipeline does not interpolate or infer latitude values for such detections. Instead, the nearest SPICE-valid row to the detected PRIMARY peak is explicitly marked in all diagnostic plots, while atmospheric structure occurring prior to that location is preserved as an optical detection without over-assigned geographic coordinates.

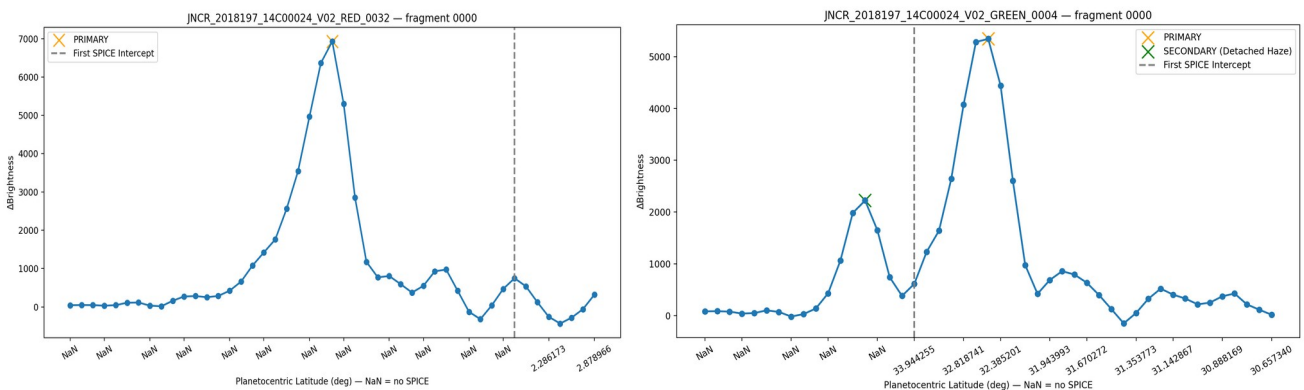
Stage 6 carries forward fragment-level detectability metadata produced in Stage 4b without modifying peak-detection logic. This design preserves a clean scientific provenance chain: Stage 4 supplies sample-level geometry, Stage 4b converts that geometry into fragment-level detectability metrics, and Stage 6 attaches those metrics to the fragment-level classification output without altering the peak-finding logic itself.

6.3 Visualization and Diagnostic Outputs

Figures 6.3 and 6.4 present representative examples illustrating both profiles: one with a single dominant limb peak and no detected secondary brightness gradient, and another in which secondary peaks are detected.

Profiles exhibiting a single dominant brightness-gradient peak are interpreted as sampling a continuous cloud deck with no resolved detached haze layer. Profiles exhibiting a secondary peak prior to the first SPICE surface intercept indicate optically detected atmospheric structure occurring above the visible limb, consistent with a detached haze morphology.

In all panels, orange markers denote the PRIMARY limb peak, while green markers denote SECONDARY peaks interpreted as candidate detached haze features. The vertical dashed line indicates the first valid SPICE surface intercept. Brightness-gradient structures occurring prior to this intercept correspond to optically detected atmospheric features for which no unique SPICE geometry can be assigned at the detected peak location.



Figures 6.3 LEFT: Representative limb-normal brightness-gradient profiles illustrating single-peak and structure classification.
Figure 6.3 RIGHT: Representative limb-normal brightness-gradient profiles illustrating double-peak structure classifications.

6.4 Known Failure Modes and Algorithmic Scope

The peak-finding algorithm operates exclusively on brightness-gradient structure and geometric constraints. It has no intrinsic semantic understanding of cloud decks versus

detached haze layers. In rare cases, a detached haze peak may be incorrectly selected as the primary peak when it rivals the cloud deck in Δ Brightness amplitude and occurs earlier along the sampling direction.

Such cases arise when:

- Two peaks have comparable prominence.
- The valley between them is shallow.
- Noise and smoothing subtly favor one feature.

Humans readily recognize these cases as a cloud deck plus detached haze system; the algorithm requires explicit geometric and SPICE-based rules to do the same. The constraints described above eliminate the majority of such failures, and residual ambiguities are flagged for manual review rather than silently excluded.

6.5 Execution

Executing Stage 6 applies automated peak detection and physical filtering to the brightness-gradient profiles produced in Stage 5, identifying candidate atmospheric features and classifying them as primary or secondary detections.

```
python src/06.peakfinder.py \  
  --indir data/JNCR_2018197_14C00024_V02/cub/stage_04_perp_samples \  
  --outdir data/JNCR_2018197_14C00024_V02/cub/stage_06_peaks \  
  --detectability-csv \  
data/JNCR_2018197_14C00024_V02/analysis/stage_04b_detectability/stage_04b_fragment_detectability.csv \  
  --smooth 5
```

Stage 6 writes per-fragment peak detection tables and diagnostic plots, while also preserving the fragment-level detectability provenance computed in Stage 4b. These products constitute the sole inputs to Stage 7, where detached haze occurrence is aggregated only across accepted fragments that also satisfy the slant-distance detectability criterion.

Stage 7 — Analysis: Latitude-Dependent Detached Haze Occurrence

Stage 7 evaluates detached-haze candidate occurrence statistically across limb fragments rather than attempting precise spatial localization at the fragment level. Fragments are included in occurrence statistics only if they satisfy the Stage 4b detectability criterion.

7.1 Limb-Crossing Latitude Assignment

For each accepted, detectability-valid limb fragment, the planetocentric latitude associated with the detected limb crossing is used for binning. Operationally, Stage 7 first attempts to assign latitude from the sampling row marked as the primary peak. If that row lacks finite

SPICE latitude, the nearest SPICE-valid row to the primary peak is used instead. This produces a limb-crossing latitude that is physically aligned with the detected limb transition and is therefore more appropriate for final occurrence statistics than the earlier first-available-latitude proxy.

For auditability and comparison with earlier exploratory analyses, the first finite SPICE latitude encountered within each fragment is still retained in the fragment-level table. Fragments lacking any finite SPICE latitude after this procedure are excluded from latitude-binned aggregation.

7.2 Binning and Occurrence Metric

Within each bin, the fraction of fragments exhibiting a secondary brightness-gradient peak is computed. This fraction represents the occurrence rate of detached haze candidates within that latitude band. Importantly, the retained fragment population includes both detached-haze detections and geometry-valid non-detections. Fragments excluded prior to aggregation are limited to cases lacking physically interpretable limb geometry or failing the Stage 04b detectability criterion. As a result, the reported occurrence metric reflects detached-haze occurrence within the retained detectable population rather than a positive-only subset.

Because earlier stages explicitly remove framelets lacking a usable limb-space transition, and Stage 7 further excludes fragments that fail the slant-distance detectability criterion, the occurrence metric reported here reflects detectability within the retained, geometrically valid fragment population rather than global atmospheric coverage. In latitude ranges where usable limb geometry is sparse or absent, low or zero occurrence values may therefore result from data exclusion rather than physical non-existence of detached haze. This conservative interpretation is maintained throughout the paper.

7.3 Results

Within the detectability-screened fragment population, detached-haze candidate occurrence increases strongly toward mid-to-higher planetocentric latitudes, approaching near-continuous occurrence above approximately 30–35° in this observation.

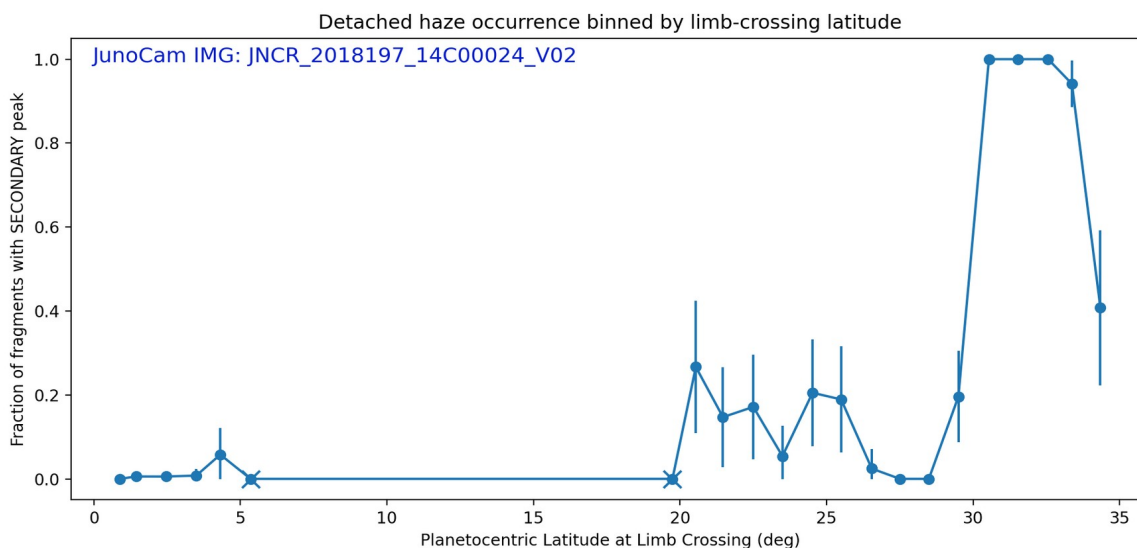


Figure 7.3 - Fraction of accepted, detectability-valid limb fragments exhibiting secondary brightness-gradient peaks (detached haze candidates), binned by limb-crossing SPICE planetocentric latitude for JunoCam IMG JNCR_2018197_14C00024_V02.

7.4 Limitations and Interpretation

Detached haze detections are not spatially localizable at their true altitude due to SPICE limitations. The reported trend therefore remains statistical rather than spatially resolved at the fragment level. However, by restricting the Stage 7 population to fragments that pass the slant-distance detectability criterion, the revised pipeline ensures that non-detections are interpreted only where detached haze would have been resolvable under the observed viewing geometry.

The observed increase in detached haze occurrence toward higher planetocentric latitude no longer reflects an undifferentiated mix of valid and non-diagnostic negatives, but instead a filtered population in which the instrument's resolving power is explicitly taken into account. For PJ14, application of this criterion retains the analyzed fragment population while strengthening the interpretive basis of the latitude-dependent trend.

7.5 Execution

Stage 7 aggregates Stage 6 fragment classifications into latitude bins to compute detached-haze occurrence as a function of limb-crossing planetocentric latitude. No new feature detection is performed; only accepted fragments that satisfy the slant-distance detectability criterion are retained.

```
python src/07.haze_analysis.py \  
  --imgdir data/JNCR_2018197_14C00024_V02 \  
  --stage6-root data/JNCR_2018197_14C00024_V02/cub/stage_06_peaks \  
  --outdir data/JNCR_2018197_14C00024_V02/analysis/stage_07 \  
  --bin-width-deg 1.0 \  
  --min-fragments-per-bin 15
```

This stage produces a fragment-level table, a latitude-binned occurrence table, and a summary visualization for the analyzed IMG, representing the terminal single-image outputs of the pipeline.

Stage 8 — Perijove-Scale Detached Haze Analysis

Because Stage 8 performs no new feature detection but does recover limb-crossing geometry from prior-stage outputs, it is presented here as a Perijove-scale aggregation of the validated Stages 1–7 pipeline. Stage 8 ingests the detectability-screened Stage 7 fragment tables and reopens the corresponding Stage 6 outputs only to recover limb-crossing latitude, longitude, and slant distance at the detected primary peak (or nearest finite SPICE fallback) for spatial aggregation.

Purpose

Stages 1–7 define a reproducible analysis pipeline operating on a single JunoCam image, producing fragment-level detections of detached haze features at limb-crossing locations. In Stage 8, this validated pipeline is applied without modification across the full set of JunoCam images acquired during a single Perijove, enabling Perijove-scale aggregation of

detached haze occurrence as a function of planetary latitude and longitude.

For this study, Stage 08 is applied to all processed JunoCam images from Perijove 14 (PJ14), with aggregation restricted to images contributing non-empty detectability-screened fragment populations.

Inputs and Aggregation Strategy

Because Stage 07 has already restricted the fragment population to accepted, detectability-valid fragments, each retained fragment contributes independently to the aggregate statistics as part of a fragment-weighted occurrence estimate rather than an image-level average. Stage 08 therefore inherits the slant-distance detectability criterion through its Stage 07 inputs rather than recomputing it at the Perijove level.

Across PJ14, Stage 01 produced 4,962 individual framelets. Following Stage 02 limb tracing and manual quality control, 1,968 framelets containing accepted, physically interpretable limb-space transitions were retained for downstream analysis, while 2,994 framelets were excluded due to absent or non-physical limb geometry, unresolved limb-space separation, or failed quality-control validation. The retained Stage 02 population was frozen into explicit QC manifest tables prior to Stage 03 processing, ensuring deterministic reproducibility and preventing interrupted reruns or partial processing failures from silently altering the analyzed fragment population.

To recover map-ready limb-crossing coordinates, Stage 8 reopens the corresponding Stage 6 per-sample CSVs and identifies the primary peak row for each retained fragment. When finite SPICE latitude, longitude, and slant distance are available at that row, those values are adopted directly; otherwise, the nearest finite SPICE-resolved row to the primary peak is used as a fallback. This procedure preserves the limb-crossing interpretation already adopted in Stage 7 while enabling longitude-latitude aggregation at Perijove scale.

Fragment aggregation is subject to several safeguards. Bins and cells are masked unless minimum fragment thresholds are met, and fragments lacking finite SPICE geometry at or near the detected limb-crossing location are excluded from spatial aggregation (but still retained in the intermediate outputs for auditability).

Execution

Stage 8 is executed once per Perijove, after all images have been processed, using the following command:

```
python src/08.perijove_analysis.py \  
  --pjdir data/PJ14 \  
  --pj-label "Perijove 14" \  
  --lat-bin-width-deg 1.0 \  
  --lon-bin-width-deg 5.0 \  
  --min-fragments-per-lat-bin 30 \  
  --min-fragments-per-img-lat-bin 8 \  

```

```
--min-fragments-per-lonlat-cell 10
```

Results

8.1 Perijove-Scale Latitude Dependence

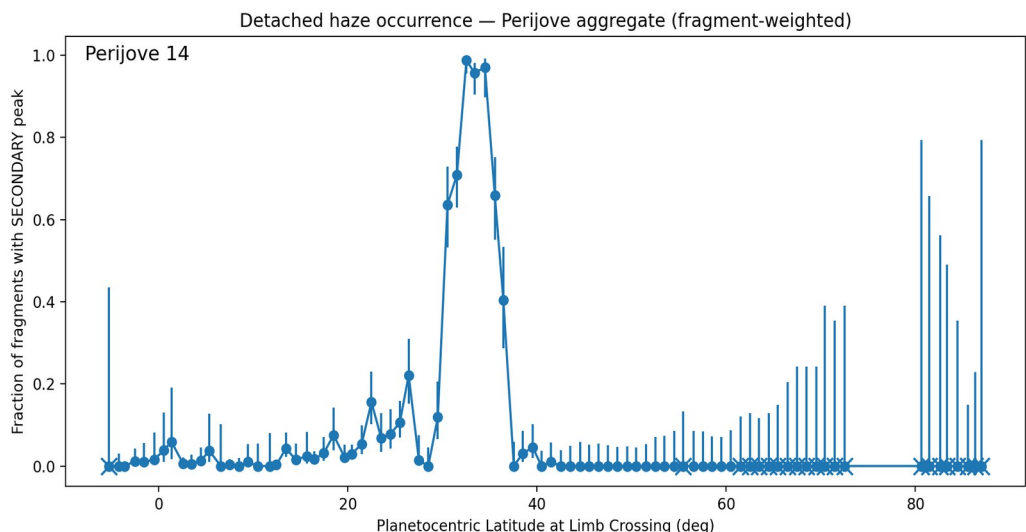
Figure 8.1 addresses the central question of this study: whether detached-haze candidate occurrence within the detectability-screened fragment population exhibits a systematic dependence on planetocentric latitude when aggregated across an entire Perijove. The fragment-weighted occurrence curve is computed exclusively from the retained detectability-valid fragment population.

The Perijove-scale occurrence curve shows a pronounced mid-latitude enhancement near $30\text{--}35^\circ$ planetocentric latitude. Below this transition, occurrence rates remain low or near zero in many latitude bins; however, low occurrence in the revised pipeline must not be interpreted automatically as physical absence of detached haze. Because Stage 8 operates exclusively on the retained detectability-valid fragment population, low occurrence may instead reflect sparse retained limb geometry, limited longitudinal sampling, or insufficient fragment support after geometric masking. Importantly, the pronounced $\sim 30\text{--}35^\circ$ enhancement persists within the well-sampled, detectability-valid subset itself, indicating that the observed mid-latitude maximum is not explained solely by observational geometry or fragment attrition.

Confidence intervals widen toward the poles as fragment counts decrease, consistent with reduced sampling density and more restrictive limb geometry at extreme latitudes. Importantly, the pronounced $\sim 30\text{--}35^\circ$ mid-latitude enhancement persists after Stage 02 QC curation, Stage 04b detectability screening, Perijove-scale aggregation, and deterministic rerun validation, indicating that the signal is not driven by small-number statistics, sparse geometry alone, or unstable denominator structure. The enhancement remains coherent across multiple independently processed images within the retained detectability-valid population. This result demonstrates that latitude-dependent detached-haze candidate occurrence emerges only when aggregating many independent limb fragments across images and remains stable after explicit geometric detectability screening.

Figure 8.1 -
Fragment-weighted
detached haze
occurrence vs.
planetocentric latitude
for Perijove 14,
computed from the
detectability-screened
Stage 7 fragment
population.

Detached haze
occurrence exhibits a
strong mid-latitude
maximum centered
near $\sim 33^\circ$, with
substantially lower
occurrence at both
equatorial and higher-
latitude bins.



8.2 Perijove-Scale Latitude Heatmap

Figure 8.2 presents a Perijove-scale heatmap of detached-haze occurrence as a function of latitude, derived from the detectability-screened fragment population retained in Stage 8. Only images contributing non-empty detectability-screened fragment populations are represented. Rather than emphasizing individual image curves, this representation highlights the concentration of occurrence within a narrow mid-latitude band. A pronounced and coherent mid-latitude enhancement appears near $\sim 30\text{--}35^\circ$ planetocentric latitude, consistent with the peak in the fragment-weighted aggregate occurrence curve. The revised detectability-screened heatmap demonstrates that this enhancement persists across multiple independently processed images rather than arising from isolated high-occurrence outliers. Regions with insufficient fragment support are masked, preserving a strict distinction between low occurrence and inadequate retained coverage.

This heatmap provides a compact visual summary of the retained detectable sample and reinforces that the dominant PJ14 signal is confined to a narrow mid-latitude regime rather than broadly distributed across all sampled latitudes. Conversely, masked or low-occurrence regions should not be interpreted as evidence for physical absence of detached haze, because Stage 8 explicitly distinguishes geometry-limited coverage from detectability-valid non-detections.

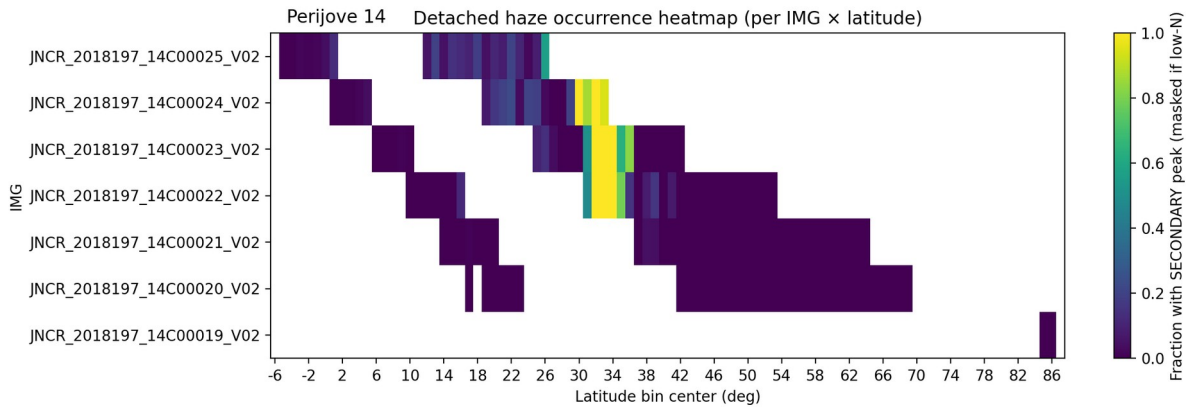


Figure 8.2 Perijove-scale heatmap of detached-haze candidate occurrence as a function of planetocentric latitude for PJ14. Each row represents an individual JunoCam image with non-empty detectability-screened fragment tables, and colors indicate fragment-weighted occurrence within latitude bins. Regions with insufficient fragment support are masked. A pronounced and coherent mid-latitude enhancement ($\sim 30\text{--}35^\circ$) persists across multiple independently processed images within the detectability-screened fragment population, consistent with the aggregate occurrence curve.

8.3 Longitude–Latitude Distribution

Figure 8.3 places the latitude-dependent results in longitudinal context by mapping detached haze occurrence onto a longitude–latitude grid for Perijove 14. This figure addresses whether the observed enhancement is globally distributed or confined to specific longitudinal sectors.

Detached haze detections cluster within limited longitude ranges at mid-latitudes, while large portions of the map remain unpopulated. These gaps reflect the inherently sparse and non-uniform longitudinal coverage imposed by the spacecraft's Perijove trajectory rather than an absence of haze. No extrapolation is performed in unsampled regions, preserving a strict distinction between observed structure and coverage limitations.

Because Stage 8 operates on the detectability-screened fragment population and retains only fragments with finite limb-crossing latitude, longitude, and slant distance, the resulting map represents where detached haze was measurably detectable under PJ14's viewing geometry, not where haze is physically absent. In the revised version of this figure, the latitude range is plotted well beyond 30° in order to retain the full mid-latitude detectability regime identified across PJ14.

Together with Figures 8.1 and 8.2, the longitude–latitude map demonstrates that the latitude-dependent enhancement is not an artifact of longitudinal averaging, while simultaneously highlighting the need for multi-Perijove aggregation to assess global longitudinal structure.

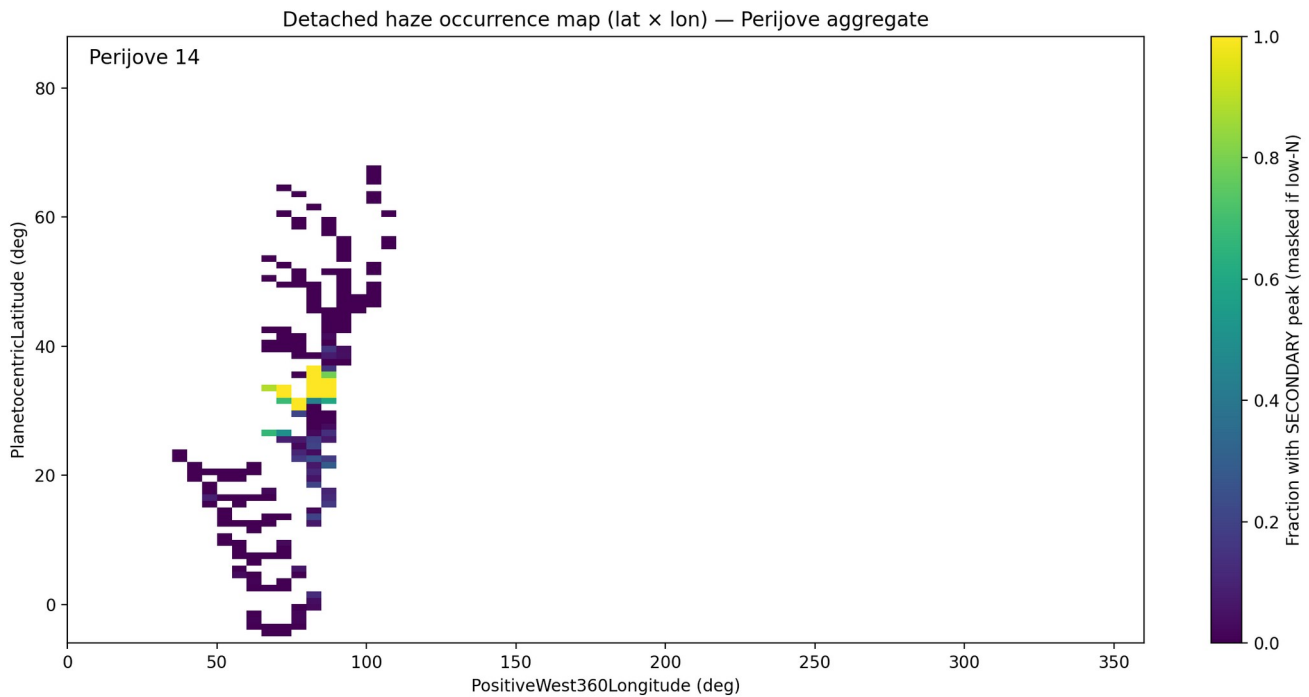


Figure 8.3. Longitude–latitude occurrence map for detached haze during Perijove 14. Colors indicate the fraction of detectability-valid fragments exhibiting **SECONDARY** peaks within each longitude–latitude cell; cells below the minimum fragment threshold are masked. The mid-latitude enhancement is retained in map space but remains coverage-limited in longitude under PJ14 viewing geometry.

8.4 Derived Physical Measurements: Peak Separation

Beyond binary detached-haze occurrence, the retained fragment population permits quantitative measurement of the apparent separation between the visible Jovian limb and the detached haze layer. Because Stage 6 identifies both **PRIMARY** (main limb) and **SECONDARY** (detached haze) brightness-gradient maxima for each retained fragment, the separation between these features can be measured directly along the limb-normal sampling coordinate.

For each retained fragment, the detector-coordinate peak separation is converted into physical distance (km) using the local SPICE-derived image scale. Applying this procedure to the Perijove 14 dataset yielded 818 retained detached-haze fragments possessing measurable **PRIMARY** and **SECONDARY** peaks, with a mean peak separation of 69.1 km

(median 68.9 km, $\sigma = 11.9$ km). The resulting distribution of peak separations is summarized in Figure 8.4:

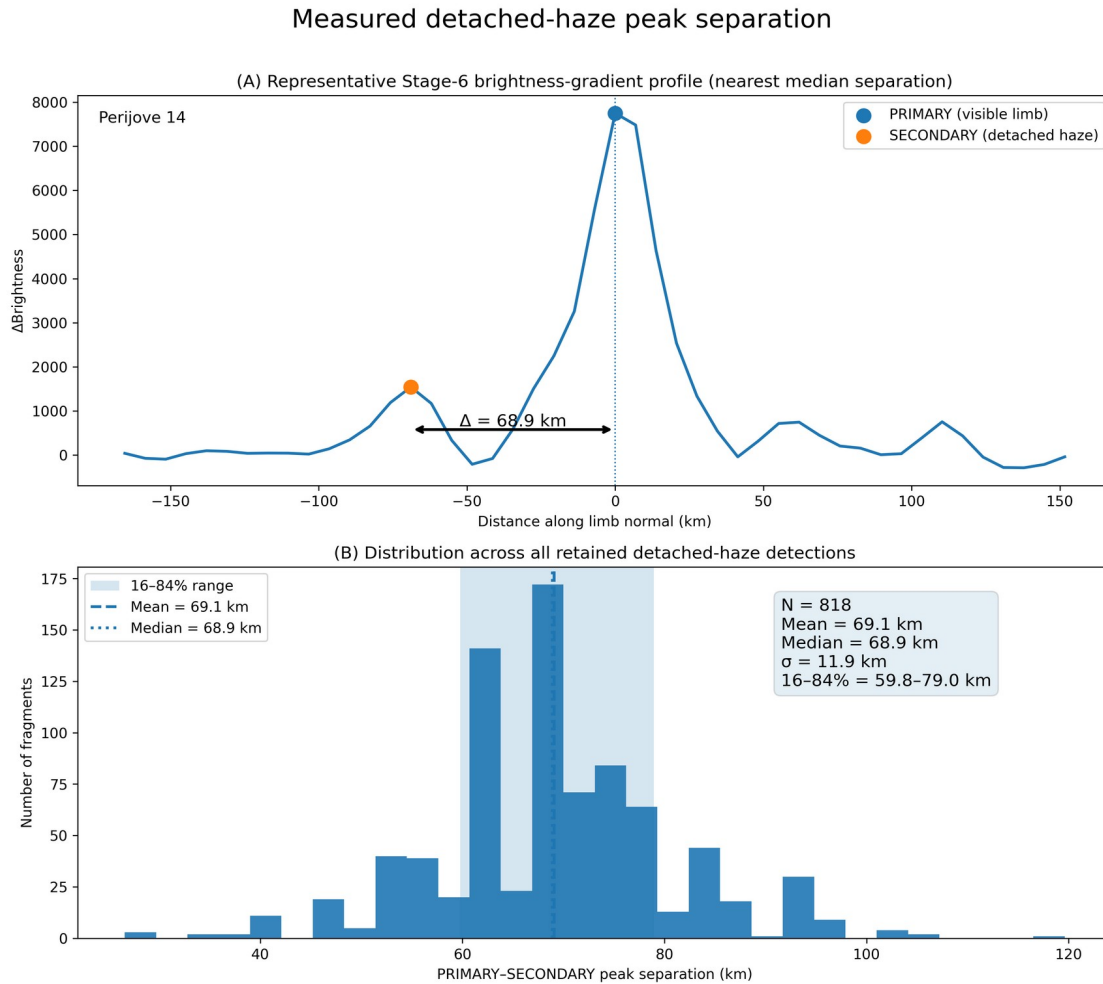


Figure 8.4. Representative detached-haze brightness-gradient profile and distribution of measured PRIMARY-SECONDARY peak separations for retained PJ14 detections. The upper panel illustrates the fragment nearest the median measured separation, while the lower panel summarizes the distribution of PRIMARY-SECONDARY separations across all retained fragments.

These measurements provide an empirical estimate of the characteristic detached-haze separation observable in PJ14 JunoCam limb imagery. Because the analysis operates on the detectability-screened fragment population established by the preceding pipeline stages, the resulting distribution represents detached haze measurable under the viewing geometry sampled during Perijove 14 and demonstrates one example of the downstream scientific products enabled by the reproducible detection pipeline.

Interpretation

Stage 8 results show that detached haze occurrence during Perijove 14 is not uniformly distributed across the sampled latitude range, exhibiting a statistically significant mid-latitude enhancement that is consistently observed across the retained detectability-valid image subset. Longitude-latitude mapping further reveals structured—but coverage-limited—spatial clustering under the Perijove’s viewing geometry. Because Stage 8 inherits

both detached-haze detections and geometry-valid non-detections from the detectability-screened Stage 7 population, low occurrence in sparsely sampled or heavily masked regions should not be interpreted as evidence for physical absence of detached haze. Instead, the revised aggregation explicitly separates geometry-limited coverage from interpretable non-detections within the retained fragment population.

Although Stage 7 outputs were generated for 28 usable PJ14 images, only 7 produced non-empty latitude-binnable fragment tables for Perijove-scale aggregation. In many excluded images, the retained Stage 6 fragment population was itself extremely small, indicating sparse usable limb geometry rather than downstream algorithmic failure. Images contributing zero Stage 7 rows were therefore treated as coverage-limited cases rather than physical non-detections.

Stage 9 — Discussion

Stage 9 interprets the Perijove-scale results of Stage 8 and introduces no additional detection or aggregation steps. A central result of the revised pipeline is the explicit separation of geometry-limited fragments from fragments that satisfy the adopted detectability criterion. As a result, low detached-haze occurrence in a given latitude range cannot automatically be interpreted as physical absence of haze, but instead must be interpreted in the context of retained limb geometry, fragment counts, longitudinal coverage, and instrument resolution. This distinction is critical because the revised pipeline preserves geometry-valid non-detections while separately excluding fragments lacking physically interpretable limb geometry. Within the retained detectability-valid population, the $\sim 30\text{--}35^\circ$ planetocentric latitude enhancement persists across multiple independently processed images and remains stable after geometric masking, QC curation, and Perijove-scale aggregation. The observed enhancement is therefore unlikely to arise solely from observational geometry or denominator structure.

Within the detectability-screened fragment population analyzed here, detached-haze candidate occurrence exhibits a pronounced and persistent mid-latitude enhancement near $30\text{--}35^\circ$ planetocentric latitude during Perijove 14. This enhancement survives QC curation, detectability filtering, geometry masking, and Perijove-scale aggregation, and appears coherently across multiple independently processed images within the retained detectable population. Because the final latitude assignment is anchored to the limb-crossing location of the primary peak, with nearest finite SPICE fallback where necessary, the reported latitude dependence remains physically tied to the detected limb transition without requiring interpolation into SPICE-undefined regions.

These results remain conservative. The analysis identifies detached-haze candidates from brightness-gradient structure and geometric consistency alone, and does not claim precise altitude retrieval at the detected haze peak. Longitude–latitude structure is likewise conditioned by sparse and nonuniform sampling, and unsampled regions are not interpreted. Future work should extend this framework across multiple Perijoves and test whether the mid-latitude enhancement persists under broader Perijove and longitudinal

sampling.

Validation analyses (Appendix B) confirm that the mid-latitude enhancement is not an artifact of fragment attrition, detectability filtering, or coverage bias. Although detectability varies with latitude, the $\sim 30\text{--}35^\circ$ region lies within a well-sampled and consistently retained subset of the data. The observed enhancement is therefore not explained by observational geometry alone and likely reflects a real atmospheric distribution.

Lastly, although JunoCam includes methane-band imaging capability, the present pipeline was developed and validated using the visible-light products analyzed here. Accordingly, detached-haze candidates are identified from brightness-gradient morphology and SPICE-constrained limb geometry rather than from methane-band opacity structure directly. Future work should test whether the mid-latitude enhancement identified here is reproduced in methane-band subsets processed under the same geometric framework.

Stage 10 — Reproducibility and Data Availability

Because all stages operate on IMG-scoped outputs and batch runners detect completed products automatically, interrupted Perijove runs may be resumed deterministically without recomputing previously validated IMG products.

Additionally, all processing code used in this study—including limb tracing, rectification, peak finding, and statistical aggregation—is publicly available in a GitHub repository at <https://github.com/kevinkell-y/juno-haze-classification>. The repository contains versioned source code, execution instructions, and directory conventions required to reproduce all figures and tables in this paper. Intermediate products generated during processing are preserved in IMG-specific directories to ensure reproducibility and facilitate independent inspection. Processed data products (rectified limb profiles, classification tables, and diagnostic figures) are derived from publicly available JunoCam images distributed by NASA's Planetary Data System. Instructions to reproduce each pipeline stage are provided inline in the Methods section. A citable archival release of this pipeline and Perijove-scale outputs is provided via Zenodo to ensure long-term reproducibility independent of repository state.

Each JunoCam cube's label and accompanying manifest file record the complete set of SPICE kernels used during initialization (LSK, PCK, SPK, CK, FK, IK, SCLK). These kernel sets were drawn from locally synchronized ISIS and Juno kernel repositories and are publicly available online. The manifest files stored alongside each .cub constitute the definitive provenance record for all geometry used in this study. After validating the pipeline on a single JunoCam image, full Perijove-scale processing may be executed using the provided batch scripts. These scripts iterate over all IMG directories and invoke Stages 01–07 sequentially, preserving per-image outputs and logs. Stage 8 is then executed once per Perijove to perform aggregate analysis.

Companion validation scripts used to audit denominator structure, detectability retention, and Perijove-scale coverage are distributed alongside the main pipeline modules.

Batch Execution:

```
# Run batch pipeline (Stages 01-07)
bash src/run_perijove_batch_stage1-2.sh

# Mandatory manual cleaning (Stage 2.3)
bash src/run_perijove_batch_stage3-7.sh
```

References

Brueshaber, S. (2021). *JunoCam Image Processing Document, Revision 2*. NASA Jet Propulsion Laboratory, Planetary Science Group Technical Report #3222.

Eichstädt, G., et al. (2021). *JunoCam Limb Geometry and Imaging Characteristics*.

Navigation and Ancillary Information Facility (NAIF). (2021). *SPICE Toolkit Documentation*.
<https://naif.jpl.nasa.gov/naif/>

Kelly, K. (2022). *Haze Detection in the Atmosphere of Jupiter Using Planetary Limb Images from JunoCam*. Internal research report, NASA Jet Propulsion Laboratory.

Wen, D. California Institute of Technology. Personal communication, 2022.

Orton, G., and Eichstädt, G, personal communication, 2025-2026.

Appendix A — Limb-Tracing Algorithm (Implementation Overview)

This appendix provides an implementation-level description of the limb-tracing procedure used in Stage 2. It is intended to document algorithmic behavior and reproducibility rather than derive formal edge-detection mathematics. The limb-tracing module (`02.trace_limb_polyline.py`) combines SPICE-registered camera geometry with grayscale gradient analysis to identify the boundary between Jupiter's illuminated atmosphere and surrounding space. The algorithm operates entirely in image-coordinate space and makes no assumption about planetary orientation, illumination direction, or limb position within the frame.

A.1 — Pre-processing

Input: SPICE-attached framelet `F.cub`

Output: Grayscale array $I(x, y)$

1. Convert `.cub` → 16-bit TIFF and 8-bit PNG using `isis2std`.
2. Read PNG and normalize pixel intensities to $[0, 1]$.
3. Apply a 3×3 Gaussian blur to suppress single-pixel noise.

A.2 — Gradient-field computation

The algorithm computes horizontal and vertical image gradients using a Sobel operator and combines them to form a gradient-magnitude field. The strongest global gradient in the framelet is used as an initial limb seed. This replaces earlier orientation-specific heuristics and ensures orbit- and geometry-agnostic limb initialization.

4. Compute Sobel derivatives: $G_x = dI/dx$, $G_y = dI/dy$
5. Compute gradient magnitude and orientation:
 $M = \sqrt{G_x^2 + G_y^2}$
 $\Theta = \arctan2(G_y, G_x)$
6. Identify global maximum of M as initial limb seed P_0 .

This replaces the earlier fixed-orientation seeding heuristic (Wen 2021) with a geometry-agnostic initialization that adapts to any illumination direction.

A.3 — Bidirectional edge propagation

7. Initialize limb polyline $L = \{P_0\}$.
8. For each propagation direction $D \in \{\text{forward, reverse}\}$:
 repeat
 • Sample candidate points Q within a narrow angular window ($\pm\theta$) around $\Theta(P_0)$.
 • Select Q maximizing local gradient magnitude $M(Q)$.
 • Append Q to L if $\text{distance}(P_0, Q) < \delta_{\text{max}}$.
 • Update orientation $\Theta(P_0) \leftarrow \Theta(Q)$; set $P_0 \leftarrow Q$ until $\text{gradient} < \tau$ or boundary reached

Typical parameters: $\theta \approx 10\text{--}15^\circ$, $\delta_{\text{max}} \approx 3$ pixels, $\tau \approx 0.1 \times M_{\text{max}}$.

Both directions are traced independently and then merged, yielding a continuous limb polyline.

A.4 — Polyline smoothing and export

9. Apply cubic spline or low-pass polynomial fit to L .
10. Save limb coordinates → `<CUB>_LIMBENDPOINTS.csv`
11. Overlay cyan markers on source image → `<CUB>_OVERLAY.png`

The resulting CSV provides ordered (x, y) positions along the detected planetary edge at sub-pixel precision. Each overlay PNG offers a rapid-visual confirmation of limb detection quality.

A.5 — Performance and false-positive control

- **Runtime:** $\approx 0.5\text{--}1$ s per 1600×128 framelet on a modern CPU.
- **Memory footprint:** < 200 MB per image.
- **False positives:** Occur when the scene lacks dark-to-bright transitions (planet fills frame). These cases are flagged during later tangent-normal sampling when no null-space data are detected.

A.6 — Scientific use

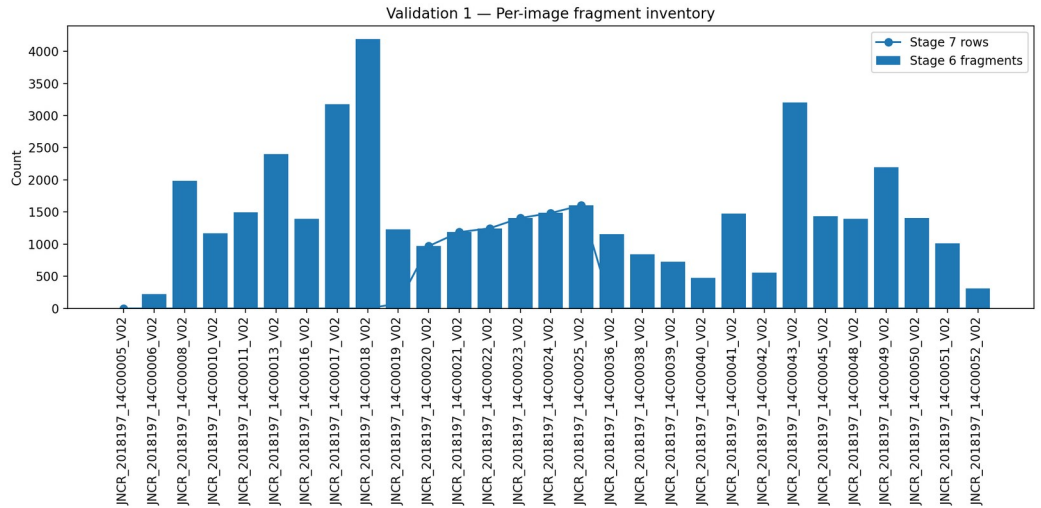
The extracted limb polylines define the instantaneous geometric curvature of Jupiter's visible atmosphere. They serve as the spatial reference for subsequent rectification and altitude mapping, enabling consistent geometric sampling for subsequent brightness-gradient analysis.

No assumption is made that the rectified limb represents a true planetary vertical; rather, rectification is performed in image-coordinate space to stabilize sampling geometry for comparative brightness-gradient analysis.

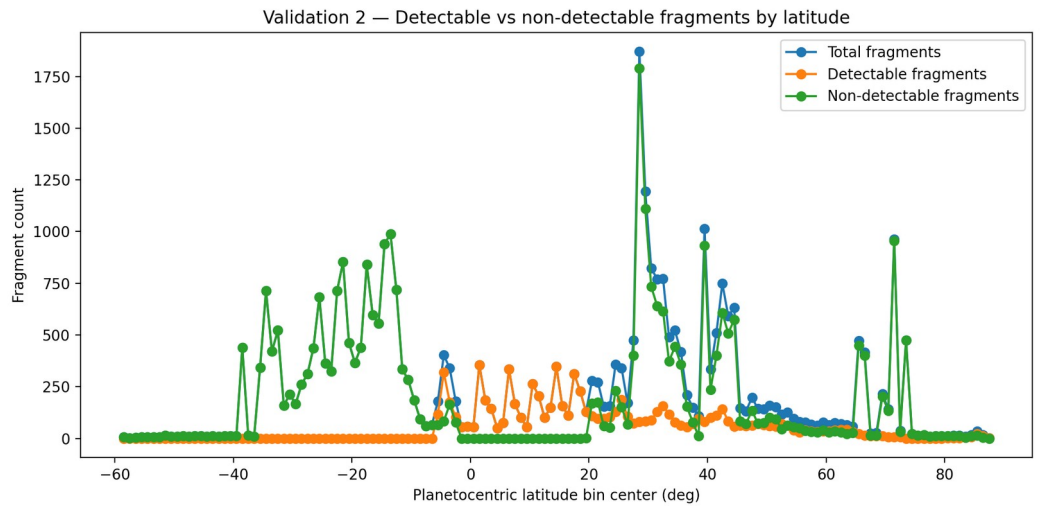
Appendix B — Validation Analyses

To verify that the observed mid-latitude enhancement is not an artifact of fragment attrition, detectability filtering, or uneven sampling, three independent validation analyses were performed at the Perijove scale. These tests evaluate (1) per-image fragment retention between Stage 6 and Stage 7, (2) latitude-dependent detectability structure across all fragments, and (3) retained coverage relative to total fragment availability. Together, these diagnostics confirm that the mid-latitude enhancement arises within a well-sampled and geometrically resolvable subset of the data.

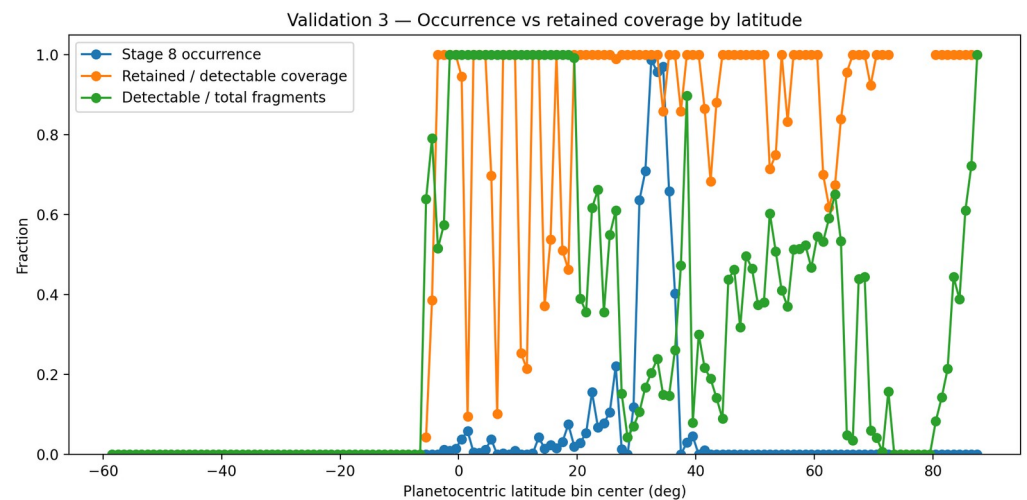
*Figure B1 —
Per-image fragment
retention
(Stage 6 vs Stage 7)*



*Figure B2 —
Detectable vs. Non-
Detectable fragments by
latitude*



*Figure B3 —
Occurrence vs. retained
coverage*



Appendix C – Environment Setup and Reproducibility Notes

Detailed operating-system requirements, ISIS3 installation procedures, SPICE kernel acquisition, and environment locking instructions are maintained in the project's github repository to ensure long-term accuracy and version control. These materials are intentionally excluded from the main manuscript, which assumes a working ISIS3 and SPICE installation, and focuses instead on the scientific pipeline and analytical methodology.